UCL ENGF0031 · SMART CLOTHING PROTOTYPE

Cadence

A wearable that senses a Parkinson's gait freeze and gently cues the next step — automatically, on the body.

ONE SENSOR. THREE JOBS.

MONITOR

Logs rest-tremor power (4–6 Hz) and daily activity from a single accelerometer.

DETECT

Spots freezing of gait using a neural network trained on real patient recordings.

RELIEVE

Fires a rhythmic vibration cue — an established, drug-free way to release the next step.

HOW IT WORKS — A TRUE CLOSED LOOP

SENSE
64 Hz accelerometer

→

DETECT
freeze? (CNN)

→

CUE
buzz the motor

The cue fires as soon as a freeze is detected, then the garment keeps watching — one sensor in, one cue out.

≈50%
of people with Parkinson's experience freezing of gait*

64 Hz
single wearable accelerometer

Drug-free
non-invasive rhythmic-cueing therapy

**Estimates vary with disease stage and assessment; commonly cited around 50%.*

In good company. Builds on ideas demonstrated by Apple Watch's Movement Disorder API, PDMonitor, STAT-ON and NextStride cueing aids — combined into one low-cost garment.

Validated on real patient recordings (Daphnet) using leave-one-subject-out testing.

REAL PATIENT DATA · LEAVE-ONE-SUBJECT-OUT

Does the freeze detector work?

FoGNet 1-D CNN · Daphnet (10 Parkinson's patients) · ankle accelerometer

0.71
of real freezes caught (sensitivity)

0.84
of walking correctly cleared (specificity)

0.78
balanced accuracy (sens + spec) / 2

CONFUSION MATRIX · n=8982

true ↓ pred →	no-freeze	freeze
no-freeze	6842 correct walk	1285 false alarm
freeze	248 MISSED freeze	607 caught freeze

VERSUS BASELINES

Approach	Sens	Spec
FoGNet (CNN)	**0.71**	**0.84**
Freeze-Index	0.002	0.994
Always 'no-freeze'	0.00	1.00

Each held-out patient is scored once, then pooled — no patient is in both train and test.

Why not "accuracy"? A model that NEVER calls a freeze scores 90.5% accuracy — yet catches zero freezes and prevents zero falls. A missed freeze can mean a fall, so sensitivity & specificity are the honest metrics here, not accuracy.

DATASET & METHOD

- 8982 four-second windows (2 s hop), one ankle accelerometer @ 64 Hz
- Freeze prevalence 9.5% (855 freeze / 8127 walk) — a rare, safety-critical event
- Leave-One-Subject-Out across all 10 patients; metrics pooled over held-out windows
- S04 & S10 have no freeze episodes → used for training only (sensitivity undefined as a test fold)

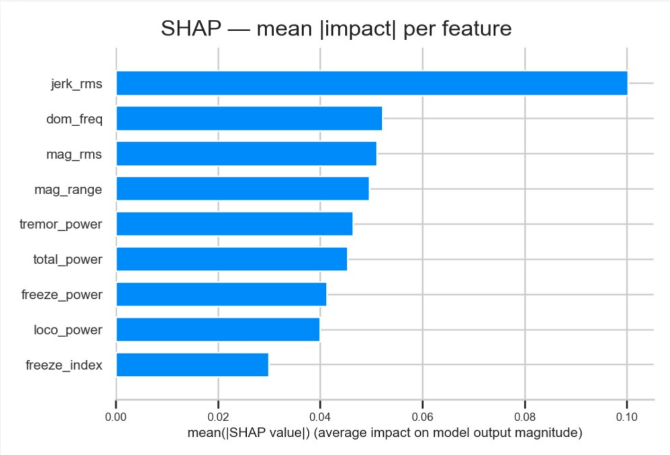
Per-patient sensitivity spans 0.45–0.98 (wide between-subject variability is expected for real Parkinson's gait; the pooled 0.71 is the headline).

MODEL INTERPRETABILITY

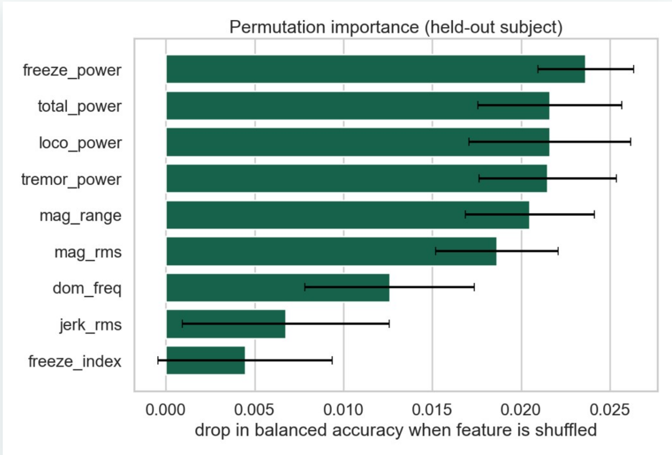
Which signals flag a freeze?

RandomForest over 9 engineered features · held-out patient S05

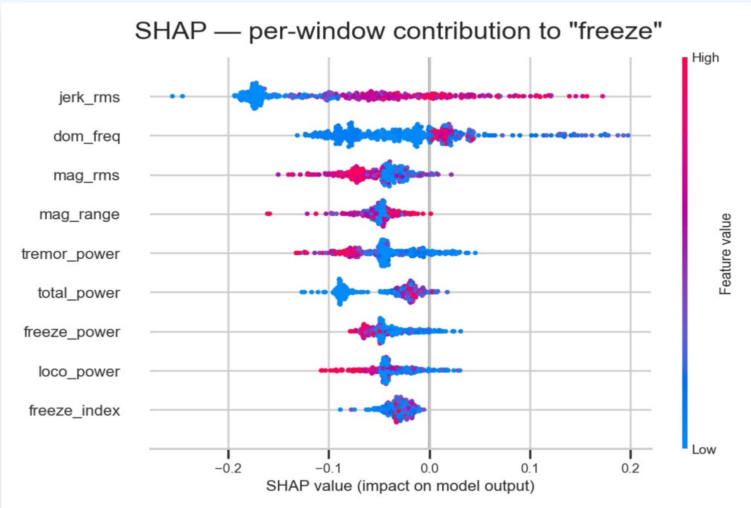
Takeaway: both SHAP and permutation importance rank the classical Freeze-Index **last** of 9 features — so the dead FI baseline is explained, not just observed. Movement dynamics (jerk, dominant frequency, magnitude) carry the signal.



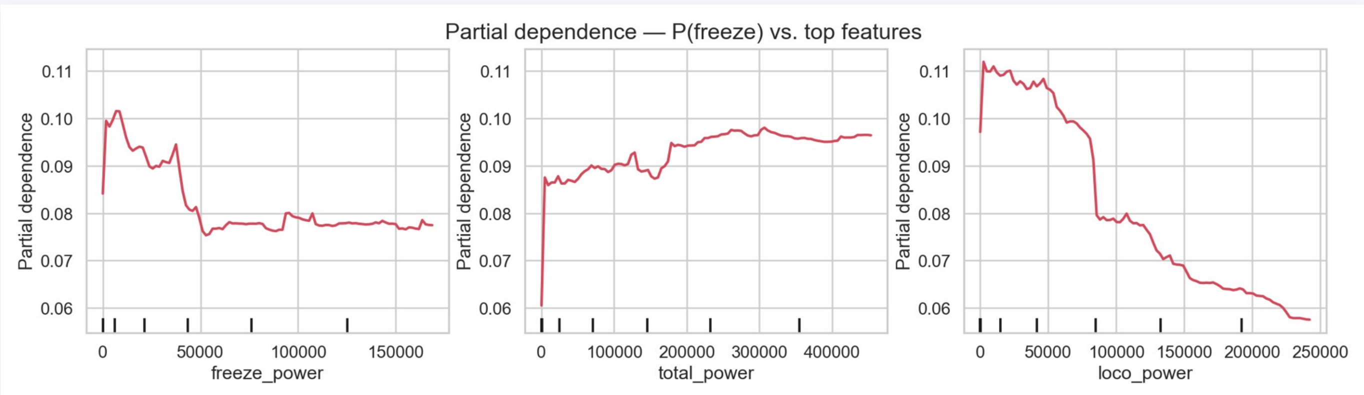
SHAP — mean impact per feature



Permutation importance — accuracy drop



SHAP beeswarm — red = high feature value



Partial dependence — P(freeze) vs. the three most influential features